Problem M3: Isospin and NN Elastic Cross Sections

Every Step, Every Clebsch–Gordan Coefficient

Midterm Preparation Notes

The Question

M3

"Similar to the example in Griffiths regarding the ratio of cross sections for $N+N \to d+\pi$, use isospin eigenstate kets to motivate why it is expected that the total cross section ratios for σ_{p+p}, σ_{p+n}, & σ_{n+n} are all one, i.e. all cross sections should be considered equal for elastic scattering in some energy regime, like high but not too high energy."

What It Wants

1. Decompose all NN states (pp, pn, nn) into isospin eigenstates using Clebsch–Gordan coefficients.
2. Write elastic scattering amplitudes in terms of the two isospin amplitudes $\mathcal{A}_0$, $\mathcal{A}_1$.
3. Show that $\mathcal{A}_0 = \mathcal{A}_1$ makes all three cross sections equal.
4. Argue physically why this holds at intermediate energies.

1 Setup: Isospin Assignments

The nucleon is an isospin-$\frac{1}{2}$ doublet:

$$|p\rangle = \left|\tfrac{1}{2}, +\tfrac{1}{2}\right\rangle, \qquad |n\rangle = \left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle \tag{1}$$

Two nucleons: $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$. The possible total isospins are $I = 1$ (triplet) and $I = 0$ (singlet). We need to express every physical two-nucleon state in terms of these eigenstates.

The Master Formula

The Clebsch–Gordan decomposition for coupling $|j_1, m_1\rangle \otimes |j_2, m_2\rangle$ into $|j, m\rangle$:

$$|j_1, j_2;\, m_1, m_2\rangle = \sum_{j=|j_1-j_2|}^{j_1+j_2} \langle j_1, j_2;\, m_1, m_2 | j,\, m_1+m_2\rangle \, |j,\, m_1+m_2\rangle \tag{2}$$

The coefficient $\langle j_1, j_2; m_1, m_2 | j, m\rangle$ is the Clebsch–Gordan (CG) coefficient. Selection rule: $m_1 + m_2 = m$.

For our case: $j_1 = j_2 = \frac{1}{2}$, each $m_i \in \{+\frac{1}{2}, -\frac{1}{2}\}$, and $j \in \{0, 1\}$.

2 Deriving Every Clebsch–Gordan Coefficient

We derive all four CG coefficients for $\frac{1}{2} \otimes \frac{1}{2}$ from scratch.

Step 1: The "stretched" state $|1, +1\rangle$

The maximum $m = +1$ can only come from $m_1 = +\frac{1}{2}$, $m_2 = +\frac{1}{2}$. There is no other combination. And the only $|j, m\rangle$ state with $m = +1$ is $|1, +1\rangle$ (since $|0, +1\rangle$ doesn't exist — you can't have $m > j$). Therefore:

CG Coefficient #1

$$|1, +1\rangle = |pp\rangle \tag{3}$$

Coefficient: $\langle \frac{1}{2}, \frac{1}{2}; +\frac{1}{2}, +\frac{1}{2} | 1, +1\rangle = 1$. Trivial — one state maps to one state.

Step 2: Obtain $|1, 0\rangle$ using the lowering operator

Apply the total lowering operator $\hat{I}_- = \hat{I}_-^{(1)} + \hat{I}_-^{(2)}$ to both sides of $|1, +1\rangle = |pp\rangle$.

Left side: Using the angular momentum lowering formula $\hat{I}_- |j, m\rangle = \sqrt{j(j+1) - m(m-1)}\, |j, m-1\rangle$:

$$\hat{I}_- |1, +1\rangle = \sqrt{1(2) - 1(0)}\, |1, 0\rangle = \sqrt{2}\, |1, 0\rangle \tag{4}$$

Right side: Apply $\hat{I}_-^{(1)} + \hat{I}_-^{(2)}$ to $|pp\rangle = |\frac{1}{2}, +\frac{1}{2}\rangle \otimes |\frac{1}{2}, +\frac{1}{2}\rangle$:

Each single-particle lowering gives: $\hat{I}_- |\frac{1}{2}, +\frac{1}{2}\rangle = \sqrt{\frac{1}{2}(\frac{3}{2}) - \frac{1}{2}(-\frac{1}{2})}\, |\frac{1}{2}, -\frac{1}{2}\rangle = 1 \cdot |\frac{1}{2}, -\frac{1}{2}\rangle$.

So:

$$(\hat{I}_-^{(1)} + \hat{I}_-^{(2)}) |pp\rangle = |np\rangle + |pn\rangle \tag{5}$$

Equating:

$$\sqrt{2}\, |1, 0\rangle = |pn\rangle + |np\rangle \tag{6}$$

CG Coefficient #2

$$|1, 0\rangle = \frac{1}{\sqrt{2}} \big(|pn\rangle + |np\rangle \big) \tag{7}$$

Coefficients: $\langle \frac{1}{2}, \frac{1}{2}; +\frac{1}{2}, -\frac{1}{2} | 1, 0\rangle = \frac{1}{\sqrt{2}}$ and $\langle \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, +\frac{1}{2} | 1, 0\rangle = \frac{1}{\sqrt{2}}$.

Step 3: Obtain $|1, -1\rangle$ by lowering again

Apply $\hat{I}_-$ to $|1, 0\rangle = \frac{1}{\sqrt{2}}(|pn\rangle + |np\rangle)$.

Left side:

$$\hat{I}_- |1, 0\rangle = \sqrt{1(2) - 0(-1)}\, |1, -1\rangle = \sqrt{2}\, |1, -1\rangle \tag{8}$$

Right side: Apply $\hat{I}_-^{(1)} + \hat{I}_-^{(2)}$ to each term:

$\hat{I}_- |pn\rangle = |nn\rangle + 0$ (can't lower n further)

$\hat{I}_- |np\rangle = 0 + |nn\rangle$ (can't lower n further)

So: $\hat{I}_- \frac{1}{\sqrt{2}}(|pn\rangle + |np\rangle) = \frac{1}{\sqrt{2}}(|nn\rangle + |nn\rangle) = \frac{2}{\sqrt{2}} |nn\rangle = \sqrt{2} |nn\rangle$

Equating:

$$\sqrt{2}\, |1, -1\rangle = \sqrt{2}\, |nn\rangle \tag{9}$$

CG Coefficient #3

$$|1,-1\rangle = |nn\rangle \tag{10}$$

Coefficient: $\langle \frac{1}{2},\frac{1}{2};-\frac{1}{2},-\frac{1}{2}|1,-1\rangle = 1$. Same logic as the stretched state.

Step 4: Obtain $|0,0\rangle$ by orthogonality

The state $|0,0\rangle$ must:

- Have $m = 0$, so it's built from $m_1 + m_2 = 0$ combinations: $|pn\rangle$ and $|np\rangle$.
- Be **orthogonal** to $|1,0\rangle = \frac{1}{\sqrt{2}}(|pn\rangle + |np\rangle)$.
- Be normalized.

Write $|0,0\rangle = \alpha|pn\rangle + \beta|np\rangle$.

Orthogonality with $|1,0\rangle$:

$$\langle 1,0|0,0\rangle = \frac{1}{\sqrt{2}}(\alpha+\beta) = 0 \quad \Longrightarrow \quad \beta = -\alpha \tag{11}$$

Normalization:

$$|\alpha|^2 + |\beta|^2 = 2|\alpha|^2 = 1 \quad \Longrightarrow \quad \alpha = \frac{1}{\sqrt{2}} \tag{12}$$

(Choosing the conventional positive sign for α.)

CG Coefficient #4

$$|0,0\rangle = \frac{1}{\sqrt{2}}\big(|pn\rangle - |np\rangle\big) \tag{13}$$

Coefficients: $\langle \frac{1}{2},\frac{1}{2};+\frac{1}{2},-\frac{1}{2}|0,0\rangle = +\frac{1}{\sqrt{2}}$ and $\langle \frac{1}{2},\frac{1}{2};-\frac{1}{2},+\frac{1}{2}|0,0\rangle = -\frac{1}{\sqrt{2}}$.
Note the minus sign — the singlet is **antisymmetric**.

Complete $\frac{1}{2} \otimes \frac{1}{2}$ Clebsch–Gordan Table (Derived)

Triplet ($I = 1$, symmetric):

$$|1,+1\rangle = |pp\rangle \tag{14}$$

$$|1,\ \ 0\rangle = \frac{1}{\sqrt{2}}\big(|pn\rangle + |np\rangle\big) \tag{15}$$

$$|1,-1\rangle = |nn\rangle \tag{16}$$

Singlet ($I = 0$, antisymmetric):

$$|0,0\rangle = \frac{1}{\sqrt{2}}\big(|pn\rangle - |np\rangle\big) \tag{17}$$

3 Inverting: Physical States in Terms of Isospin Eigenstates

For the scattering problem, we need the **inverse**: express $|pp\rangle$, $|pn\rangle$, $|nn\rangle$ in terms of $|I, I_3\rangle$.

From the results above, $|pp\rangle$ and $|nn\rangle$ are already pure eigenstates:

$$|pp\rangle = |1, +1\rangle \tag{18}$$

$$|nn\rangle = |1, -1\rangle \tag{19}$$

For $|pn\rangle$, solve from the two equations:

$$|1, 0\rangle = \frac{1}{\sqrt{2}}|pn\rangle + \frac{1}{\sqrt{2}}|np\rangle \tag{20}$$

$$|0, 0\rangle = \frac{1}{\sqrt{2}}|pn\rangle - \frac{1}{\sqrt{2}}|np\rangle \tag{21}$$

Add them:

$$|1, 0\rangle + |0, 0\rangle = \frac{2}{\sqrt{2}}|pn\rangle = \sqrt{2}\,|pn\rangle \tag{22}$$

Therefore:

$$\boxed{|pn\rangle = \frac{1}{\sqrt{2}}|1, 0\rangle + \frac{1}{\sqrt{2}}|0, 0\rangle} \tag{23}$$

Verification

Check normalization: $\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} + \frac{1}{2} = 1$. ✓

Check orthogonality: $\langle pp|pn\rangle = 0$ since pp is pure $|1, +1\rangle$ and pn has no $|1, +1\rangle$ component. ✓

4 The Griffiths Example: $NN \to d + \pi$

Final state isospin

- Deuteron: $I_d = 0$, $I_{3,d} = 0$
- Pion: $I_\pi = 1$

The final state is $|I_d = 0\rangle \otimes |I_\pi = 1, I_{3,\pi}\rangle$. Since coupling $0 \otimes 1 = 1$, this is **pure** $I = 1$.

Proving $0 \otimes 1 = 1$ (trivial table)

For $j_1 = 0$, the only value of m_1 is 0. The selection rule $m_1 + m_2 = m$ gives $m_2 = m$. The only possible total j is $j = j_2$ (since $|j_1 - j_2| = j_2$ and $j_1 + j_2 = j_2$). So:

$$|j_1 = 0, j_2;\ 0, m_2\rangle = 1 \cdot |j = j_2,\ m = m_2\rangle \tag{24}$$

The CG coefficient is 1. There is nothing to look up — coupling with zero changes nothing.

So the three final states are:

$$|d + \pi^+\rangle = |0,0\rangle \otimes |1,+1\rangle = |1,+1\rangle \tag{25}$$
$$|d + \pi^0\rangle = |0,0\rangle \otimes |1,\ 0\rangle = |1,\ 0\rangle \tag{26}$$
$$|d + \pi^-\rangle = |0,0\rangle \otimes |1,-1\rangle = |1,-1\rangle \tag{27}$$

All pure $I = 1$. Only the $I = 1$ component of the initial state contributes.

Scattering amplitudes

Since the strong interaction conserves isospin, $\hat{T}$ is diagonal in I:

$$\langle I', I_3'|\hat{T}|I, I_3\rangle = \delta_{II'}\delta_{I_3 I_3'}\,\mathcal{A}_I \tag{28}$$

(a) $pp \to d + \pi^+$: Initial $= |1,+1\rangle$, Final $= |1,+1\rangle$:

$$\mathcal{M}_a = \langle 1,+1|\hat{T}|1,+1\rangle = \mathcal{A}_1 \tag{29}$$

(b) $pn \to d + \pi^0$: Initial $= \frac{1}{\sqrt{2}}|1,0\rangle + \frac{1}{\sqrt{2}}|0,0\rangle$, Final $= |1,0\rangle$:

$$\mathcal{M}_b = \langle 1,0|\hat{T}\left(\frac{1}{\sqrt{2}}|1,0\rangle + \frac{1}{\sqrt{2}}|0,0\rangle\right) \tag{30}$$
$$= \frac{1}{\sqrt{2}}\underbrace{\langle 1,0|\hat{T}|1,0\rangle}_{=\mathcal{A}_1} + \frac{1}{\sqrt{2}}\underbrace{\langle 1,0|\hat{T}|0,0\rangle}_{=0\ (\text{isospin conserved},\ I\neq I')} \tag{31}$$
$$= \frac{1}{\sqrt{2}}\mathcal{A}_1 \tag{32}$$

(c) $nn \to d + \pi^-$: Initial $= |1,-1\rangle$, Final $= |1,-1\rangle$:

$$\mathcal{M}_c = \langle 1,-1|\hat{T}|1,-1\rangle = \mathcal{A}_1 \tag{33}$$

Cross section ratios

$\sigma \propto |\mathcal{M}|^2$:

$$\sigma_a : \sigma_b : \sigma_c = |\mathcal{A}_1|^2 : \frac{1}{2}|\mathcal{A}_1|^2 : |\mathcal{A}_1|^2 = 2 : 1 : 2 \tag{34}$$

Griffiths Result: $NN \to d\pi$

$$\boxed{\sigma(pp \to d\pi^+) : \sigma(pn \to d\pi^0) : \sigma(nn \to d\pi^-) = 2 : 1 : 2} \tag{35}$$

Exact prediction from isospin alone. Only one amplitude ($\mathcal{A}_1$) contributes.

5 The M3 Problem: $NN \to NN$ Elastic Scattering

Key difference from Griffiths

For elastic scattering, the final state is also two nucleons. Both $I = 0$ and $I = 1$ channels can contribute, so we have **two** independent amplitudes.

Amplitude for $pp \to pp$

Initial state from (18): $|pp\rangle = |1, +1\rangle$.
Final state: also $|pp\rangle = |1, +1\rangle$.

$$\mathcal{M}(pp \to pp) = \langle 1, +1|\hat{T}|1, +1\rangle = \mathcal{A}_1 \tag{36}$$

$$\boxed{\sigma_{pp} \propto |\mathcal{A}_1|^2} \tag{37}$$

Amplitude for $nn \to nn$

Initial state from (19): $|nn\rangle = |1, -1\rangle$.
Final state: also $|nn\rangle = |1, -1\rangle$.

$$\mathcal{M}(nn \to nn) = \langle 1, -1|\hat{T}|1, -1\rangle = \mathcal{A}_1 \tag{38}$$

$$\boxed{\sigma_{nn} \propto |\mathcal{A}_1|^2} \tag{39}$$

Immediate conclusion

$\sigma_{pp} = \sigma_{nn}$ **exactly**, by isospin symmetry alone, at any energy. Both are pure $I = 1$ states related by a rotation in isospin space. No assumptions about the amplitudes needed.

Amplitude for $pn \to pn$

Initial state from (23): $|pn\rangle = \frac{1}{\sqrt{2}}|1, 0\rangle + \frac{1}{\sqrt{2}}|0, 0\rangle$.
Final state: also $|pn\rangle = \frac{1}{\sqrt{2}}|1, 0\rangle + \frac{1}{\sqrt{2}}|0, 0\rangle$.

$$\mathcal{M}(pn \to pn) = \left(\frac{1}{\sqrt{2}}\langle 1, 0| + \frac{1}{\sqrt{2}}\langle 0, 0|\right) \hat{T} \left(\frac{1}{\sqrt{2}}|1, 0\rangle + \frac{1}{\sqrt{2}}|0, 0\rangle\right) \tag{40}$$

Expand all four terms:

$$= \frac{1}{2}\underbrace{\langle 1, 0|\hat{T}|1, 0\rangle}_{=\mathcal{A}_1} + \frac{1}{2}\underbrace{\langle 1, 0|\hat{T}|0, 0\rangle}_{=0} + \frac{1}{2}\underbrace{\langle 0, 0|\hat{T}|1, 0\rangle}_{=0} + \frac{1}{2}\underbrace{\langle 0, 0|\hat{T}|0, 0\rangle}_{=\mathcal{A}_0} \tag{41}$$

The cross terms vanish because $\hat{T}$ conserves isospin: $\langle I'|\hat{T}|I\rangle = 0$ when $I' \neq I$.

$$\mathcal{M}(pn \to pn) = \frac{1}{2}\mathcal{A}_1 + \frac{1}{2}\mathcal{A}_0 \tag{42}$$

$$\boxed{\sigma_{pn} \propto \left|\frac{\mathcal{A}_0 + \mathcal{A}_1}{2}\right|^2 = \frac{1}{4}|\mathcal{A}_0 + \mathcal{A}_1|^2} \tag{43}$$

General Cross Section Expressions

$$\sigma_{pp} \propto |\mathcal{A}_1|^2 \tag{44}$$

$$\sigma_{pn} \propto \frac{1}{4}|\mathcal{A}_0 + \mathcal{A}_1|^2 \tag{45}$$

$$\sigma_{nn} \propto |\mathcal{A}_1|^2 \tag{46}$$

where $\mathcal{A}_0$ and $\mathcal{A}_1$ are complex scalars (scattering matrix elements in the $I = 0$ and $I = 1$ channels). In general, $\sigma_{pn} \neq \sigma_{pp}$.

6 The Equality Condition: $\sigma_{pp} = \sigma_{pn} = \sigma_{nn}$

When does $\sigma_{pn} = \sigma_{pp}$?

We need:

$$\frac{1}{4}|\mathcal{A}_0 + \mathcal{A}_1|^2 = |\mathcal{A}_1|^2 \tag{47}$$

If $\mathcal{A}_0 = \mathcal{A}_1 \equiv \mathcal{A}$:

$$\sigma_{pn} \propto \frac{1}{4}|\mathcal{A} + \mathcal{A}|^2 = \frac{1}{4}|2\mathcal{A}|^2 = \frac{4|\mathcal{A}|^2}{4} = |\mathcal{A}|^2 \tag{48}$$

And:

$$\sigma_{pp} = \sigma_{nn} \propto |\mathcal{A}|^2 \tag{49}$$

All three equal. ✓

The 1:1:1 Result

When $\mathcal{A}_0 = \mathcal{A}_1$:

$$\boxed{\sigma_{pp} : \sigma_{pn} : \sigma_{nn} = 1 : 1 : 1} \tag{50}$$

Physical argument: why $\mathcal{A}_0 \approx \mathcal{A}_1$ at "high but not too high" energies

Three Energy Regimes

Low energy (resonance region): Scattering can proceed through specific resonances with definite isospin. If an $I = 1$ resonance exists at some energy but no $I = 0$ resonance (or vice versa), then $\mathcal{A}_1 \neq \mathcal{A}_0$ and the cross sections differ. Analogous to πN scattering at $\sqrt{s} = 1232$ MeV where the $\Delta(1232)$ resonance ($I = 3/2$) dominates and gives $\sigma(\pi^+ p)/\sigma(\pi^- p) = 3$.

Intermediate energy (above resonances, below electroweak): This is the "high but not too high" regime. The scattering becomes diffractive, dominated by Pomeron exchange. The Pomeron carries vacuum quantum numbers ($I = 0$) and is flavor-blind — it scatters $I = 0$ and $I = 1$ NN states identically. Physically, the nucleons act as "black disks" whose geometric cross sections don't depend on whether they're protons or neutrons. The u/d mass difference (~ 2 MeV) is negligible compared to the momentum scales, so the interaction is effectively charge-independent.
Result: $\mathcal{A}_0 \approx \mathcal{A}_1$, giving $\sigma_{pp} \approx \sigma_{pn} \approx \sigma_{nn}$.

Very high energy: Electroweak effects become relevant. The Coulomb interaction (which distinguishes pp from nn from pn) introduces corrections. Isospin is no longer exact. The 1:1:1 equality breaks down.

7 Comparison: $NN \to d\pi$ vs. $NN \to NN$

	$NN \to d + \pi$ **(Griffiths)**	$NN \to NN$ **(this problem)**
Final state isospin	Pure $I = 1$ (proven: $0 \otimes 1 = 1$, CG coeff $= 1$)	Both $I = 0$ and $I = 1$ possible
Number of amplitudes	1 ($\mathcal{A}_1$ only)	2 ($\mathcal{A}_0$ and $\mathcal{A}_1$)
pp amplitude	$\mathcal{A}_1$	$\mathcal{A}_1$
pn amplitude	$\frac{1}{\sqrt{2}}\mathcal{A}_1$	$\frac{1}{2}(\mathcal{A}_0 + \mathcal{A}_1)$
nn amplitude	$\mathcal{A}_1$	$\mathcal{A}_1$
Cross section ratios	$2 : 1 : 2$ (exact, any energy)	$1 : 1 : 1$ only if $\mathcal{A}_0 = \mathcal{A}_1$ (requires dynamics)

The Takeaway

- Isospin alone gives $\sigma_{pp} = \sigma_{nn}$ exactly for elastic scattering.
- Isospin alone does **not** predict σ_{pn} relative to σ_{pp} — that requires knowing $\mathcal{A}_0/\mathcal{A}_1$.
- The physical argument that scattering becomes flavor-blind at intermediate energies ($\mathcal{A}_0 \approx \mathcal{A}_1$) is what completes the 1:1:1 prediction.
- For $NN \to d\pi$, the ratios $2 : 1 : 2$ are exact because only one isospin channel contributes — no dynamics needed.

Symmetry *constrains* but doesn't always *determine*. Sometimes you need dynamics too.

Summary of Every CG Coefficient Used

Coefficient	Value	How derived	Where used
$\langle \frac{1}{2}, \frac{1}{2}; +\frac{1}{2}, +\frac{1}{2} \vert 1, +1 \rangle$	1	Stretched state (unique)	$\vert pp \rangle = \vert 1, +1 \rangle$
$\langle \frac{1}{2}, \frac{1}{2}; +\frac{1}{2}, -\frac{1}{2} \vert 1, 0 \rangle$	$\frac{1}{\sqrt{2}}$	Lowering operator on $\vert 1, +1 \rangle$	$\vert pn \rangle$ decomposition
$\langle \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, +\frac{1}{2} \vert 1, 0 \rangle$	$\frac{1}{\sqrt{2}}$	Lowering operator on $\vert 1, +1 \rangle$	$\vert pn \rangle$ decomposition
$\langle \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, -\frac{1}{2} \vert 1, -1 \rangle$	1	Lowering operator on $\vert 1, 0 \rangle$	$\vert nn \rangle = \vert 1, -1 \rangle$
$\langle \frac{1}{2}, \frac{1}{2}; +\frac{1}{2}, -\frac{1}{2} \vert 0, 0 \rangle$	$+\frac{1}{\sqrt{2}}$	Orthogonality with $\vert 1, 0 \rangle$	$\vert pn \rangle$ decomposition
$\langle \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, +\frac{1}{2} \vert 0, 0 \rangle$	$-\frac{1}{\sqrt{2}}$	Orthogonality with $\vert 1, 0 \rangle$	$\vert pn \rangle$ decomposition
$\langle 0, 1; 0, m \vert 1, m \rangle$	1	Trivial: $0 \otimes j = j$	$d + \pi$ final state

Every coefficient is either derived from the lowering operator, from orthogonality, or is trivial (stretched state or coupling with zero). Nothing was looked up without proof.